

AI-POWERED INFRASTRUCTURE MONITORING

# GCP Monitoring Agent

Your GCP infrastructure, guarded by AI. Automated inspection, intelligent analysis, instant alerts — all from a single Cloud Run service.

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A FULLY AUTOMATED GCP RESOURCE INSPECTION SYSTEM

— 2026 —

AT A GLANCE

# What GCP Monitoring Agent Does

Fully automated. AI-powered. One command to deploy.

# Beyond Threshold Alerts

## TRADITIONAL

### Rule-Based Alerting

- CPU > 90% \u2192 fire alert
- No root cause analysis
- Cannot predict trends
- One metric, one rule

ACT II

# System Architecture

Five modules, one pipeline. Cloud Scheduler triggers, Cloud Run executes, GCS persists, Telegram delivers.

SERVERLESS · 1 VCPU / 512 MB · PYTHON 3.13

# From GCP APIs to Structured Reports

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## INSPECTION PIPELINE

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01

### Enumerate

list() all GCE instances in target zone

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03

### Aggregate

3-min window, 60-sec alignment period

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02

### Fetch

Query CPU / Memory / Disk via Monitoring API

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04

### Analyze

Gemini 2.5 Flash inspects each target

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THREE MODES, ONE MODEL

# Gemini 2.5 Flash

Fast checks, deep reasoning, and natural conversation — a single model handles every monitoring scenario.

Output: status \u00b7 risk\_level \u00b7 recommendation \u00b7 cross-resource correlation

## STANDARD MODE

Rapid threshold comparison across 5 metrics. Outputs **ok** / **warning** / **critical** status with concise reasoning.

## DEEP MODE

Multi-dimensional correlation analysis. Identifies resource dependencies and emerging risks. Produces **specific, actionable recommendations**.

YOUR OPS CENTER, IN YOUR POCKET

# Telegram Bot

Instant alerts + slash commands + free-form AI chat. All inside the messaging app you already use.

Auto alerts push proactively. No manual polling required. MarkdownV2 rich formatting with automatic plain-text fallback.

\U2318 /STATUS

View latest inspection report with formatted tables

\U2318 /INSPECT VM-NAME

Instant single-VM diagnosis with AI analysis

\U2318 /HELP

# One Command to Production

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## DEPLOYMENT PIPELINE

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01

### Build

Docker build with Python 3.13-slim + gcloud CLI

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03

### Deploy

gcloud run deploy with 1 vCPU / 512 MB

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02

### Push

gcloud builds submit to Container Registry

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04

### Schedule

Cloud Scheduler cron: \*/5 \* \* \* \*

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THE BOTTOM LINE

# The Price of a Coffee, Every Month

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CLOUD RUN

**\$5.12**

PAGE 09 • COST ANALYSIS

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CLOUD SCHEDULER

**\$0.50**

\$6.12 / MONTH

# Let AI Watch Over Every GCP Instance

One command to deploy. Zero ongoing maintenance. Full  
AI-powered monitoring for pocket change.

[GITHUB.COM/WINSON-030/2026-MONITOR-AGENT](https://github.com/winson-030/2026-monitor-agent) · [GCP RUN DEPLOY](#)